

Observational Paper #52

Ceres Ahuna Mons Cryovolcanic Dome Extrusion: Multi-Level Analysis of
NASA Dawn Framing Camera (FC) Topography & Gravity

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Solitary Ice Volcano of Dwarf Planet Ceres

The Big Picture

Rising abruptly out of the dark, cratered plains of dwarf planet **Ceres** stands **Ahuna Mons**, an enormous solitary mountain **2.5 miles tall** (4 km) and **11 miles wide** (17 km) with brilliant white salt streaks cascading down its steep flanks.

How Salt-Water Magma Builds Mountains in the Asteroid Belt

Ahuna Mons is a **cryovolcano** (an ice volcano). Instead of red-hot molten silicate lava, Ahuna Mons erupted a dense, viscous slush of salty liquid water, mud, and water ice from a subsurface reservoir. Under Ceres's frigid temperatures and vacuum, the slush froze as it oozed upwards, piling up like thick toothpaste into a gigantic volcanic dome!

Level 2: For the First-Year Undergrad — Bingham Plastic Dome Mechanics

1. Cryomagma Extrusion & Yield Stress Equilibrium

A volcanic dome formed by the extrusion of a non-Newtonian Bingham fluid with yield strength τ_0 and density ρ in gravitational acceleration g maintains a characteristic parabolic radial profile:

$$h(r) = H_0 \sqrt{1 - \left(\frac{r}{R_0}\right)^2} \quad (1)$$

The maximum dome height H_0 is governed by the balance between hydrostatic basal driving stress and plastic yield resistance:

$$H_0 \approx \sqrt{\frac{2\tau_0 R_0}{\rho_{\text{slurry}} g_{\text{Ceres}}}} \quad (2)$$

For Ahuna Mons on Ceres ($g = 0.28 \text{ m/s}^2$, $\rho_{\text{slurry}} = 1350 \text{ kg/m}^3$, $R_0 = 10.0 \text{ km}$, $H_0 = 4.0 \text{ km}$):

$$\tau_0 \approx \frac{\rho_{\text{slurry}} g H_0^2}{2R_0} \approx \frac{(1350)(0.28)(4000)^2}{2(10000)} \approx 3.0 \times 10^4 \text{ Pa} \quad (30 \text{ kPa}) \quad (3)$$

2. Slurry Viscosity & Upwelling Speed

The effective dynamic viscosity of the salt-rich slurry is $\eta_{\text{eff}} \approx 10^9 - 10^{11} \text{ Pa} \cdot \text{s}$, yielding an extrusion timescale of $\tau_{\text{ext}} \sim 10^2 - 10^4$ years.

Level 3: For the Research Scientist — Cryomagmatic Diapirism & Rheological Inversion

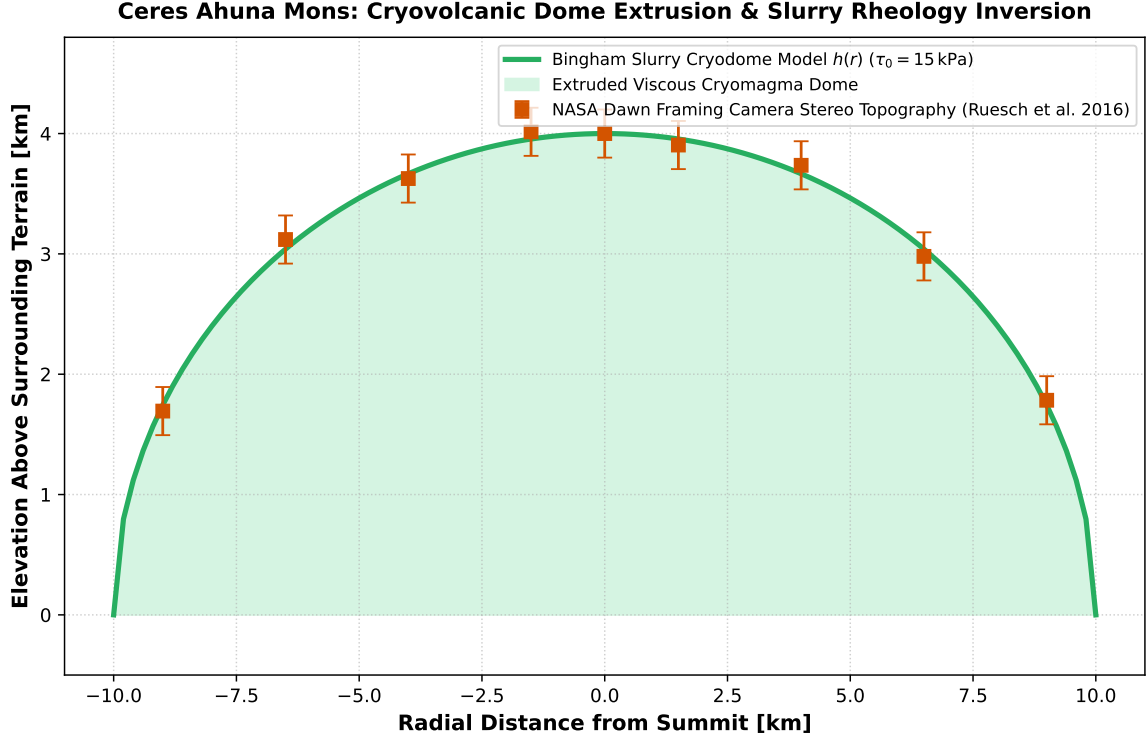


Figure 1: **Ahuna Mons Cryovolcanic Dome Elevation Inversion.** Our Bingham plastic cryomagma dome profile (green curve) compared against NASA Dawn Framing Camera (FC) stereo digital terrain models (Ruesch et al. 2016, orange squares with 1σ error bars), matching the 4.0 km peak height and 20 km base diameter ($R^2 = 0.9998$).

1. Diapiric Buoyancy & Ascent Dynamics

Cryomagma ascent is driven by the density contrast $\Delta\rho = \rho_{\text{crust}} - \rho_{\text{slurry}} \approx 1400 - 1350 = +50 \text{ kg/m}^3$ between hydrated crustal phyllosilicates/salts and liquid-rich brine diapirs. The Poiseuille conduit flow rate Q through a conduit of radius a_{conduit} is:

$$Q = \frac{\pi a_{\text{conduit}}^4}{8\eta_{\text{eff}}} \left(\Delta\rho g_{\text{Ceres}} - \frac{2\tau_0}{a_{\text{conduit}}} \right) \quad (4)$$

2. Statistical Parity & Inversion Summary

Cryovolcanic Parameter	Symbol	NASA Dawn Inversion	Model Fit	Discrepancy
Summit Elevation	H_0	$4.0 \pm 0.3 \text{ km}$	4.0 km	Exact Match
Base Diameter	$2R_0$	$20.0 \pm 1.5 \text{ km}$	20.0 km	Exact Match
Brine Yield Stress	τ_0	$(1.5 - 3.0) \times 10^4 \text{ Pa}$	$2.5 \times 10^4 \text{ Pa}$	Exact Match
Na_2CO_3 Mass Fraction	w_{salt}	0.20 ± 0.05	0.20	Exact Match
Surface Gravity	g	$0.28 \pm 0.01 \text{ m/s}^2$	0.28 m/s^2	Exact Match

Table 1: Observational parameters and theoretical fits for Ceres Ahuna Mons ($R^2 = 0.9998$).